

A01 AIR QUALITY | P

Intent : Provide a basic level of indoor air quality that contributes to the health and well-being of building users.

Summary : This WELL feature requires projects to provide acceptable air quality levels, as determined by public health authorities.

Issue : Exposure to air pollutants, such as Volatile Organic Compounds (VOCs), ozone, particulate matter, carbon monoxide and others has been shown to increase the risk of respiratory and cardiovascular diseases, in addition to causing thousands of cancer deaths annually.¹ Inhaling pollutants presents symptoms including headaches, dry throat, eye irritation and runny nose that may later develop into extreme health outcomes, such as asthma attacks and cancer.^{2–4} In addition, radon exposure is the second leading cause of lung cancer, after tobacco use.⁵ Therefore, it is important to define indoor air quality levels that minimize risk to human health.

Solutions : The World Health Organization (WHO) and other regulatory bodies, such as the U.S. Environmental Protection Agency (EPA) identify a list of “criteria” air pollutants. They have established permissible levels for such criteria pollutants based on epidemiological studies that show the relationships between concentrations of these pollutants, duration of exposure and health risks. Achieving the goal of clean indoor air as defined by permissible levels, requires the joint efforts of both professionals and building users in the implementation of adequate approaches. Indoor air quality can be properly managed through different features listed in the WELL Air concept, including source control strategies, passive and active building design and operation strategies and human behavior interventions. Effective mechanical ventilation is particularly effective at bringing radon below acceptable thresholds.^{6,7}

PART 1 MEET THRESHOLDS FOR PARTICULATE MATTER

For All Spaces except Commercial Kitchen Spaces & Industrial:

Option 1: Acceptable thresholds

The following thresholds are met in occupiable spaces:

a. PM_{2.5}: 15 µg/m³ or lower.⁸

b.

PM₁₀: 50 µg/m³ or lower.⁹

OR

Option 2: Modified thresholds in polluted regions

Note: Projects pursuing this strategy are limited in WELL Certification level to Gold regardless of total points achieved.

For buildings where the annual average outdoor PM_{2.5} level is 35 µg/m³ or higher, the following thresholds are met:

a. PM_{2.5}: 25 µg/m³ or lower.¹⁰

b. PM₁₀: 50 µg/m³ or lower.¹⁰

OR

Option 3: Dynamic thresholds in polluted regions

Note: Projects pursuing this strategy are limited in WELL Certification level to Silver regardless of total points achieved.

For buildings where the annual average outdoor PM_{2.5} level is 35 µg/m³ or higher, the following thresholds are met:

a. PM_{2.5} less than or equal to 30% of the 24- or 48-hour average of outdoor levels on the day(s) of performance testing.

b. PM₁₀ less than or equal to 30% of the 24- or 48-hour average of outdoor levels on the day(s) of performance testing.

For Commercial Kitchen Spaces & Industrial:

Option 1: Acceptable thresholds

The following threshold is met:

a. PM_{2.5}: 35 µg/m³ or lower.⁸

OR

Option 2: Dynamic thresholds in polluted regions

Note: Projects pursuing this strategy are limited in WELL Certification level to Silver regardless of total points achieved.

For buildings where the annual average outdoor PM_{2.5} level is 35 µg/m³ or higher, the following thresholds are met:

a. PM_{2.5} equal to 30% of the 24- or 48-hour average of outdoor levels on the day(s) of performance testing.

b. PM₁₀ equal to 30% of the 24- or 48-hour average of outdoor levels on the day(s) of performance testing.

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details.

The World Health Organization’s Global Urban Ambient Air Pollution Database may be consulted to view outdoor air quality levels, available at <https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database>.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 MEET THRESHOLDS FOR ORGANIC GASES

For All Spaces:

Option 1: Laboratory-based VOC tests

The following thresholds are met in occupiable spaces:

- Benzene (CAS 71-43-2): 10 µg/m³ or lower.¹¹
- Formaldehyde (CAS 50-00-0): 50 µg/m³ or lower.¹²
- Toluene (CAS 108-88-3): 300 µg/m³ or lower.¹³

OR

Option 2: TVOC continuous monitoring

The following threshold is met in occupiable spaces:

- Total VOC: 500 µg/m³ or lower.

Note : Projects undergoing recertification which were previously awarded Feature A08 must consider all data collected since the previous (re)certification.

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 3 MEET THRESHOLDS FOR INORGANIC GASES

For All Spaces except Commercial Kitchen Spaces & Industrial:

The following thresholds are met in occupiable spaces:

- Carbon monoxide: 10 mg/m³ [9 ppm] or lower.⁸
- Ozone: 100 µg/m³ [51 ppb] or lower.¹⁰

For Commercial Kitchen Spaces & Industrial:

The following thresholds are met:

- Carbon monoxide: 34 mg/m³ [30 ppm] or lower.¹⁴
-

Ozone: 100 µg/m³ [51 ppb] or lower.¹⁰

Note :

Multifamily residential projects may achieve WELL Certification at the Bronze or Silver level without testing in dwelling units, but cannot achieve Gold or Platinum without testing in dwelling units. See Sampling Rates for Multifamily Residential in the WELL Performance Verification Guidebook for further details.

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 4 MEET THRESHOLDS FOR RADON

For All Spaces:

Option 1: Radon testing

The following threshold is met on the lowest regularly occupiable floor:

- Radon: 0.15 Bq/L [4 pCi/L] or lower, as tested by a professional demonstrated not to have a conflict of interest with the WELL project. One test is conducted per 2300 m² of regularly occupied space.

OR

Option 2: Mechanical ventilation

For regularly occupied spaces at or below grade, the following requirement is met:

- All regularly occupied spaces at or below grade meet Feature A03, Part 1, Option 1.

OR

Option 3: Above-grade

One of the following requirements are met in regularly occupied spaces:

- Is completely located on or above the third floor of the building.
- Is constructed with raised-pier foundations (e.g., without a solid perimeter wall) and all mechanical equipment is elevated off the ground.

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 5 MEASURE AIR PARAMETERS

For All Spaces except Dwelling Units:

The following pollutants are monitored in occupiable spaces at intervals no longer than once per year, and the results are submitted annually through the WELL digital platform. To determine the required sample locations and quantity, refer to the WELL Performance Verification Guidebook.

- a. PM_{2.5}
- b. PM₁₀.
- c. One of the following:
 1. Total VOC.
 2. Benzene, Formaldehyde, Toluene.
- d. Carbon Monoxide.
- e. Ozone.

Note : Projects are not required to follow the device requirements or test methods described in the Performance Verification Guidebook.

REFERENCES

1. Franklin BA, Brook R, Arden Pope C. Air pollution and cardiovascular disease. *Curr Probl Cardiol*. 2015;40(5):207-238. doi:10.1016/j.cpcardiol.2015.01.003
2. Joshi S. The sick building syndrome. *Indian J Occup Environ Med*. 2008;12(2):61. doi:10.4103/0019-5278.43262
3. Selgrade MK, Plopper CG, Gilmour MI, Conolly RB, Foos BSP. Assessing the health effects and risks associated with children's inhalation exposures - Asthma and allergy. *J Toxicol Environ Heal - Part A Curr Issues*. 2008;71(3):196-207. doi:10.1080/15287390701597897
4. U.S. Environmental Protection Agency. Indoor Air Pollution: An Introduction for Health Professionals. <https://www.epa.gov/indoor-air-quality-iaq/indoor-air-pollution-introduction-health-professionals>.
5. World Health Organization. WHO Handbook on Indoor Radon: A Public Health Perspective. Presented at the: 2009. https://www.who.int/ionizing_radiation/env/9789241547673/en/.
6. Yang S, Goyette Pernot J, Hager Jörin C, Niculita-Hirzel H, Perret V, Licina D. Radon Investigation in 650 Energy Efficient Dwellings in Western Switzerland: Impact of Energy Renovation and Building Characteristics. *Atmosphere (Basel)*. 2019;10(12):777. doi:10.3390/atmos10120777
7. Hadlich DE, Grimsrud DT. Radon in Institutional Buildings: The Impacts of Conservation Strategies. *Radon Institutional Build Impacts*. 1991;(1990).
8. U.S. Environmental Protection Agency. NAAQS Table. <https://www.epa.gov/criteria-air-pollutants/naaqs-table>.
9. European Commission. Proposal for a directive of the European Parliament and of the Council on ambient air quality and cleaner air for Europe - COM(2005) 447 Final. *Off J Eur Union*. 2005;0183:1-67.
10. World Health Organization. Air Quality Guidelines for Particulate Matter, Ozone, Nitrogen Dioxide and Sulfur Dioxide. Geneva, Switzerland; 2005. http://apps.who.int/iris/bitstream/handle/10665/69477/WHO_SDE_PHE_OEH_06.02_eng.pdf;jsessionid=CD44465Fsequence=1.
11. French Agency for Food Environmental and Occupational Health and Safety. ANSES's List of Indoor Air Quality Guideline Values.; 2018. https://www.anses.fr/fr/system/files/Tableau_VGAI_Juillet2018EN.pdf.
12. Health Canada. Residential Indoor Air Quality Guideline: Formaldehyde.; 2006. <https://www.canada.ca/content/dam/canada/health-canada/migration/healthy-canadians/publications/healthy-living-vie-saine/formaldehyde/alt/formaldehyde-eng.pdf>.
13. California Office of Environmental Health Hazard Assessment. Acute, 8-hour and Chronic Reference Exposure Level (REL) Summary. <https://oehha.ca.gov/air/general-info/oehha-acute-8-hour-and-chronic-reference-exposure-level-rel-summary>. Published 2019.
14. Health and Safety Executive. EH40/2005 Workplace Exposure Limits.; 2011. <http://www.hse.gov.uk/pUbns/priced/eh40.pdf>.

A02 SMOKE-FREE ENVIRONMENT | P

Intent : Deter smoking, minimize occupant exposure to secondhand smoke and reduce smoke pollution.

Summary : This WELL feature requires projects to ban indoor smoking and ban or restrict outdoor smoking within its boundaries.

Issue : Exposure to tobacco smoke has continued to detrimentally affect the health of both smokers and those exposed to secondhand smoke. Ingredients in cigarettes form over 7,000 compounds when burned, of which at least 69 are known to be carcinogenic.¹ As a result, the average life expectancy of a smoker is 10 years less than that of a nonsmoker.² Furthermore, secondhand smoke exposes nonsmokers to the same contaminants, increasing the number of people subject to health risks from smoking. The thirdhand smoke (residual chemicals left on indoor surfaces by tobacco smoke) clings to walls, furniture, clothing, bedding, carpets and other surfaces, long after smoking has occurred. Emerging evidence suggests that there are serious health consequences associated with exposure to thirdhand tobacco smoke,³ secondhand marijuana smoke and emissions from e-cigarettes.^{4,5} Health issues associated with tobacco smoke include asthma attacks, respiratory infections, coronary heart disease, stroke, lung cancer and sudden infant death syndrome.⁶ Despite these adverse impacts on health, 80% of the global population lives in countries not protected by 100% smoke-free regulations.⁷

Solutions : There is no safe and acceptable level of cigarette smoke exposure.⁸ Therefore, the only way to protect people from secondhand and thirdhand smoke is to implement a 100% smoke-free environment.⁸⁻¹⁰ In order to prevent intrusion of cigarette smoke from the outdoors, projects must also take steps to ensure that smoking is not allowed in the vicinity of building entrances, operable windows and building air intakes.¹¹

PART 1 PROHIBIT INDOOR SMOKING

For All Spaces:

The following requirement is met:

- a. Smoking and the use of e-cigarettes is prohibited in interior spaces within the project boundary.¹²

PART 2 PROHIBIT OUTDOOR SMOKING

For All Spaces:

Option 1: No smoking signage

The following requirements are met:

- a.
 - Clear and visible permanent signage prohibiting smoking and vaping is located within 10 ft of all functional building entrances, operable windows and building air intakes that open to any occupiable outdoor area.
 - Clear and visible permanent signage describing the hazards of smoking is located in all outdoor areas designated for smoking and vaping.

OR -----

Option 2: No applicable outdoor spaces

The following are not present within the project boundary:

- a. Any occupiable outdoor areas (e.g., decks, patios, balconies, rooftops, walkways)
- b. Functional building entrances.

REFERENCES

1. National Cancer Institute. Harms of Cigarette Smoking and Health Benefits of Quitting. <https://www.cancer.gov/about-cancer/causes-prevention/risk/tobacco/cessation-fact-sheet>.
2. Jha P, Ramasundarahettige C, Landsman V, et al. 21st-Century Hazards of Smoking and Benefits of Cessation in the United States. *N Engl J Med*. 2013;368(4):341-350. doi:10.1056/NEJMsa1211128
3. Matt GE, Quintana PJE, Destailats H, et al. Thirdhand tobacco smoke: Emerging evidence and arguments for a multidisciplinary research agenda. *Environ Health Perspect*. 2011;119(9):1218-1226. doi:10.1289/ehp.1103500
4. Wang X, Derakhshandeh R, Liu J, et al. One Minute of Marijuana Secondhand Smoke Exposure Substantially. *J Am Heart Assoc*. 2016;5(8):1-11. doi:10.1161/JAHA.116.003858
5. Mello S, Bigman CA, Sanders-Jackson A, Tan ASL. Perceived harm of secondhand electronic cigarette vapors and policy support to restrict public vaping: Results from a national survey of US adults. *Nicotine Tob Res*. 2016;18(5):686-693. doi:10.1093/ntr/ntv232
6. U.S. Department of Health and Human Services. The Health Consequences of Smoking: 50 Years of Progress: A Report of the Surgeon General. Rockville, MD; 2014. https://www.ncbi.nlm.nih.gov/books/NBK179276/pdf/Bookshelf_NBK179276.pdf.
7. World Health Organization. WHO Report on the Global Tobacco Epidemic, 2017 - Monitoring Tobacco Use and Prevention Policies. Geneva, Switzerland; 2017. <http://apps.who.int/iris/bitstream/handle/10665/255874/9789241512824-eng.pdf?sequence=1%0Ahttps://www.world-heart-federation.org/wp-content/uploads/2017/07/WHO-Report-on-the-global-tobacco-epidemic-2017-EMBARGOED.pdf%0Ahttp://apps.who.int/iris/bitstream/1>.
8. International Agency for Research on Cancer. Evaluating the Effectiveness of Smoke-Free Policies. Geneva, Switzerland; 2009. <https://www.iarc.fr/wp-content/uploads/2018/07/handbook13.pdf>.

9. Gan Q, Hammond SK, Jiang Y, Yang Y, Hu TW. Effectiveness of a smoke-free policy in lowering secondhand smoke concentrations in offices in China. *J Occup Environ Med*. 2008;50(5):570-575. doi:10.1097/JOM.0b013e3181638640
10. MacNaughton P, Adamkiewicz G, Arku RE, Vallarino J, Levy DE. The impact of a smoke-free policy on environmental tobacco smoke exposure in public housing developments. *Sci Total Environ*. 2016;557-558:676-680. doi:10.1016/j.scitotenv.2016.03.110
11. Kaufman P, Zhang B, Bondy SJ, Klepeis N, Ferrence R. Not just "a few wisps": Real-time measurement of tobacco smoke at entrances to office buildings. *Tob Control*. 2011;20(3):212-218. doi:10.1136/tc.2010.041277
12. New York State Department of Health. NYS Public Health Law, Article 13-E: Regulation of Smoking in Public and Work Places. New York, NY; 2014. <https://www.nysenate.gov/legislation/laws/PBH/A13-E>.
13. U.S. Green Building Council. LEED BD+C: New Construction | v4 - LEED v4. 2018. https://www.usgbc.org/sites/default/files/LEED_v4_BDC_07.2.18_current.pdf.
14. Hammond D, Fong GT, McNeill A, Borland R, Cummings KM. Effectiveness of cigarette warning labels in informing smokers about the risks of smoking: Findings from the International Tobacco Control (ITC) Four Country Survey. *Tob Control*. 2006;15(SUPPL. 3):iii19-25. doi:10.1136/tc.2005.012294

A03 VENTILATION DESIGN | P

Intent : Minimize indoor air quality issues through the provision of adequate ventilation.

Summary : This WELL feature requires projects to bring in fresh air from the outside through mechanical and/or natural means in order to dilute human- and product-generated air pollutants.

Issue : Poorly ventilated spaces contribute to symptoms, such as headache, fatigue, dizziness, nausea, cough, sneezing, shortness of breath and eye, nose, throat and skin irritation. This is collectively called sick building syndrome (SBS).^{1,2} Poor ventilation is also linked to increased rates of employee absences, higher operational costs for businesses and decreased productivity in students.^{3,4} One U.S.-based study reported that the sick leave attributable to insufficient provision of fresh air in buildings is estimated to be 35% of total absenteeism.⁵ Therefore, the economic costs of SBS in under-ventilated buildings are significant and far exceed the energy-related cost savings.⁶⁻⁸

Solutions : Many indoor and outdoor sources of air pollution emit particulate matter and volatile organic compounds (VOCs) that can cause discomfort and trigger asthma and eye, nose and throat irritation. In order to maintain healthy indoor environments and acceptable air quality for building users, it is necessary to provide sufficient ventilation.^{9,10} In addition to proper HVAC system design, mechanically ventilated projects need to perform regular system maintenance, since inadequate maintenance is associated with reduced ventilation performance and a deterioration in indoor air quality and thermal conditions.¹¹ For naturally ventilated spaces, it is necessary to ensure that adequate outdoor air quality and noise levels are met.

PART 1 ENSURE ADEQUATE VENTILATION

For All Spaces:

Option 1: Mechanically ventilated spaces

For mechanically ventilated buildings, one of the following requirements is met:

- a. Newly installed ventilation systems are designed to meet the supply and exhaust rates set in one or more of the following ventilation guidelines, which must describe ventilation rates for at least 90% of the project area. The ventilation system is scheduled to be tested and balanced after project occupancy:
 1. ASHRAE 62.1-2010 or any more recent versions (Ventilation Rate Procedure or IAQ Procedure).¹²
 2. ASHRAE 62.2-2016.¹³
 3. EN 16798-1.¹⁴
 4. AS 1668.2-2012 or any more recent version.¹⁵
 5. CIBSE Guide A: Environmental Design, version 2007 or any more recent version.¹⁶
- b. Existing ventilation systems have been tested and balanced to meet supply and exhaust rates set in one or more ventilation guidelines listed above within the last five years.

OR

Option 2: Naturally ventilated spaces

For naturally ventilated buildings with no mechanical ventilation, the following requirements are met:

- a. One or more of the following design criteria, which must describe ventilation rates for at least 90% of the project area:
 1. Natural Ventilation Procedure in ASHRAE 62.1-2010 or any more recent version.¹²
 2. CIBSE AM10: Natural Ventilation in Non-Domestic Buildings (2005 or any more recent version) Section 2.4 – Natural ventilation strategies and Chapter 4 – Design Calculations.¹⁷
 3. AS 1668.4-2012 or any more recent version.¹⁵
 4. Any reference in Option 1, which describes natural ventilation procedures.
- b. Vents and windows used to meet the ventilation requirements in one of the standards mentioned above are permanently open or have controls to prevent their closure during periods of occupancy. (Operable windows not used in ventilation calculations may be user operated.)
- c. Outdoor air meets the following thresholds as an average for the previous year:
 1. PM_{2.5} less than 15 µg/m³.¹⁸
 2. PM₁₀ less than 30 µg/m³.¹⁸

OR

Option 3: Naturally ventilated spaces in areas with elevated particulate matter

Note: Projects pursuing this strategy are limited in WELL Certification level to Silver regardless of total points achieved.

For naturally ventilated buildings with no mechanical ventilation, the following requirements are met:

- a. One or more of the following design criteria, which must describe ventilation rates for at least 90% of the project area:

1. Natural Ventilation Procedure in ASHRAE 62.1-2010 or any more recent version.¹²
 2. CIBSE AM10: Natural Ventilation in Non-Domestic Buildings (2005 or any more recent version) Section 2.4 – Natural ventilation strategies and Chapter 4 – Design Calculations.¹⁷
 3. AS 1668.4-2012 or any more recent version.¹⁵
 4. Any reference in Option 1, which describes natural ventilation procedures.
- b. Vents and windows used to meet the ventilation requirements in one of the standards mentioned above are permanently open or have controls to prevent their closure during periods of occupancy. (Operable windows not used in ventilation calculations may be user operated.)
- c. Outdoor air meets the following thresholds as an average for the previous year:
1. PM_{2.5} less than 35 µg/m³.¹⁸
 2. PM₁₀ less than 70 µg/m³.¹⁸

OR

Option 4: Ventilation monitoring

One of the following carbon dioxide thresholds is met in occupiable spaces:

- a. 900 ppm or less.
- b. Not more than 500 ppm above outdoor levels.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

Note :

Mixed mode projects must meet the requirements set for naturally ventilated spaces when mechanical system is not in use and the requirements for mechanical ventilation when the mechanical system is in use. The project must use mechanical ventilation when outdoor PM_{2.5} levels are above 15 µg/m³, based on a measuring device within 4 km of the project.

REFERENCES

1. Wargocki P, Wyon DP, Sundell J, Clausen G, Fanger PO. The effects of outdoor air supply rate in an office on perceived air quality, sick building syndrome (SBS) symptoms and productivity. *Indoor Air*. 2000;10(4):222-236. <https://onlinelibrary.wiley.com/doi/epdf/10.1034/j.1600-0668.2000.010004222.x>.
2. Daisey JM, Angell WJ, Apte MG. Indoor air quality, ventilation and health symptoms in schools: An analysis of existing information. *Indoor Air*. 2003;13(1):53-64. doi:10.1034/j.1600-0668.2003.00153.x
3. Haverinen-Shaughnessy U, Moschandreas DJ, Shaughnessy RJ. Association between substandard classroom ventilation rates and students' academic achievement. *Indoor Air*. 2011;21(2):121-131. doi:10.1111/j.1600-0668.2010.00686.x
4. Chan WR, Parthasarathy S, Fisk WJ, Mckone TE. Estimated effect of ventilation and filtration on chronic health risks in U.S. offices, schools, and retail stores. *Indoor Air*. 2016;26(2):331-343. doi:10.1111/ina.12189
5. Milton DK, Glencross PM, Walters MD. Risk of sick leave associated with outdoor air supply rate, humidification, and occupant complaints. *Indoor Air*. 2000;10(4):212-221. https://buildequinox.com/files/iaq/milton_vent_sick_rates.pdf.
6. Redlich CA, Sparer J, Cullen MR. Sick-building syndrome. *Lancet (London, England)*. 1997;349(9057):1013-1016. <https://www.ncbi.nlm.nih.gov/pubmed/9100639>.
7. Fisk WJ. Estimates of improved productivity and health from better indoor environments. *Indoor Air*. 1997;7(3):158-172. doi:10.1111/j.1600-0668.1997.t01-1-00002.x
8. Fisk WJ. How IEQ affects health, productivity. *ASHRAE*. 2002;44(5).
9. Sundell J, Levin H, Nazaroff WW, et al. Ventilation rates and health: Multidisciplinary review of the scientific literature. *Indoor Air*. 2011;21(3):191-204. doi:10.1111/j.1600-0668.2010.00703.x
10. Carrer P, Wargocki P, Fanetti A, et al. What does the scientific literature tell us about the ventilation-health relationship in public and residential buildings? *Build Environ*. 2015;94(P1):273-286. doi:10.1016/j.buildenv.2015.08.011
11. Hanssen SO. HVAC - The importance of clean intake section and dry air filter in cold climate. *Indoor Air, Suppl*. 2004;14(SUPPL. 7):195-201. doi:10.1111/j.1600-0668.2004.00288.x
12. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 62.1-2010: Ventilation for Acceptable Indoor Air Quality.; 2010.

13. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 62.2-2016: Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings. <https://www.ashrae.org/technical-resources/standards-and-guidelines/standards-addenda/addenda-to-standard-62-2-2016>. Published 2016.
14. European Committee for Standardization. EN 16798-1:2019 - Ventilation for Buildings. 2019. https://standards.cen.eu/dyn/www/f?p=204:110:0:::FSP_PROJECT,FSP_ORG_ID:41425,6138&cs=11EDD0CE838BCEF1A1EFA39A24B6C9890.
15. Standards Australia. AS 1668-2012 The Use of Ventilation and Air-Conditioning in Buildings. Sydney; 2012.
16. Chartered Institution of Building Services Engineers. CIBSE Guide A: Environmental Design. Chartered Institution of Building Services Engineers; 2007. doi:10.1016/B978-0-240-81224-3.00016-9
17. Chartered Institution of Building Services Engineers. AM 10 Natural Ventilation in Non Domestic Buildings.; 2005. <https://www.cibse.org/knowledge/knowledge-items/detail?id=a0q20000008I7m2AAC>.
18. World Health Organization. Air Quality Guidelines for Particulate Matter, Ozone, Nitrogen Dioxide and Sulfur Dioxide. Geneva, Switzerland; 2005. http://apps.who.int/iris/bitstream/handle/10665/69477/WHO_SDE_PHE_OEH_06.02_eng.pdf;jsessionid=CD44465Fsequence=1.

A04 CONSTRUCTION POLLUTION MANAGEMENT | P

Intent : Minimize the introduction of construction-related pollutants into indoor air, remediate construction-related indoor air contamination for human health and protect building products from degradation.

Summary : This WELL feature requires projects to protect indoor air quality during building construction and renovation through a combination of strategies, such as duct protection, moisture and dust management, filter replacement, and proper equipment selection.

Issue : Air pollution originating from construction and demolition works can impact the health and quality of life of people working and residing nearby and is associated with increased mortality, due to chronic obstructive pulmonary disease.¹ Increased emissions of PM₁₀ in the vicinity of construction sites are positively correlated with people's suffering from respiratory diseases.² Building construction and renovations are primarily a source of coarse particulate matter, but they are also a source of airborne ultrafine particles.³

Solutions : The indoor air quality levels can be compromised during building construction. Care should be taken during construction to prevent dust intrusion and/or to clear the space of dust, chemical vapors and other debris to avoid significant aggravation of indoor air quality. Air duct protection, moisture and dust management, filter replacement and proper equipment selection are strategies that improve indoor air quality, by limiting the exposure to an intense contamination period.⁴

PART 1 MITIGATE CONSTRUCTION POLLUTION

For All Spaces:

Option 1: Manage construction pollution

For construction occurring after enrollment or the start of subscription, the following requirements are met:

- a. Ducts are maintained per one of the below:
 1. Ducts are sealed and protected from possible contamination during construction.⁵
 2. Ducts are cleaned prior to installing registers, grills and diffusers.
- b. If permanently installed ventilation system is operating during construction, filters must meet the following:
 1. Media filters with an average removal efficiency of $\geq 70\%$ for particles 3-10 μm in size (e.g., MERV 8, M5 or ePM10 70%) are used to filter return air.⁶
 2. All filters are replaced prior to occupancy.⁵
- c. The project implements the following moisture and dust management procedures:
 1. Carpets, acoustical ceiling panels, fabric wall coverings, insulation, upholstery and furnishings and other absorptive materials are stored separately in a designated area protected from moisture damage.⁵
 2. All active areas of work are isolated from other spaces by sealed doorways or windows or through the use of temporary barriers.⁵
 3. Walk-off mats are used at entryways to reduce the transfer of dirt and pollutants.⁵
 4. Saws and similar tools use dust guards or collectors to capture generated dust.⁵

OR

Option 2: No construction activities

The following requirement is met:

- a. The project has not undergone construction activities after enrollment.

REFERENCES

1. Bergdahl IA, Torén K, Eriksson K, et al. Increased mortality in COPD among construction workers exposed to inorganic dust. *Eur Respir J*. 2004;23(3):402-406. doi:10.1183/09031936.04.00034304
2. Ravinder Singh, Kafeel Ahmad DCJ and MSK. Impact of Air Quality on Human Health In The Vicinity of Construction Sites in Delhi-NCR. *Int J Eng Res Appl*. 2014;4(8):18-26. http://www.ijera.com/papers/Vol4_issue8/Version 4/C48041826.pdf.
3. Kumar P, Mulheron M, Som C. Release of ultrafine particles from three simulated building processes. *J Nanoparticle Res*. 2012;14(4):771. doi:10.1007/s11051-012-0771-2
4. Indoor Air Quality Program. Construction and renovation generated pollutants in occupied buildings. <https://www.mass.gov/service-details/constructionrenovation-generated-pollutants-in-occupied-buildings>.
5. U.S. Green Building Council. LEED BD+C: New Construction | v4 - LEED v4. 2018. <https://www.usgbc.org/resources/leed-v4-building-design-and-construction-current-version> <https://www.usgbc.org/resources/leed-v4-building-design-and-construction-current-version>.
6. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 52.2-2017: Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size. 2017. https://www.techstreet.com/ashrae/standards/ashrae-52-2-2017?product_id=1942059.

A05 ENHANCED AIR QUALITY | O (MAX : 4 PT)

Intent : Encourage and recognize buildings with enhanced levels of indoor air quality that promote the health and well-being of people.

Summary : This WELL feature requires projects to go beyond current guidelines to provide enhanced air quality levels that have been linked to improved human health and performance.^{1,2}

Issue : The quality of the air people breathe indoors directly impacts their health and well-being and constitutes one of the most important aspects of buildings that can support human health. Researchers have also identified a clear relationship between indoor air quality and human productivity in buildings.³ An average of 10% of productivity loss could be attributable to health issues related to poor indoor air quality in office buildings.⁴ There is also an emerging body of evidence that air pollution can disrupt physical and cognitive development in children.⁵ Studies have also shown that air pollution contributes to the large global burden of respiratory and allergic diseases, as well as the premature deaths of adults and children.⁶ The premature mortality rate could be reduced by up to 15%, if PM₁₀ is reduced from 70 to 20 µg/m³.⁷ As a result, enhanced air quality is positively correlated with improved health, cognitive and physical development, higher incomes and better economic performance.^{1,2}

Solutions : Indoor air quality can be properly managed primarily through source control strategies, passive and active building design and operation strategies and human behavior intervention. High levels of indoor air quality require both professionals and building users to collaborate in the implementation of adequate approaches.

PART 1 MEET ENHANCED THRESHOLDS FOR PARTICULATE MATTER (MAX : 2 PT)

For All Spaces:

The following requirement is met:

- a. Projects comply with the thresholds specified in the table below:

Tier	Particulate Matter Thresholds	Points
1	PM _{2.5} : 12 µg/m ³ or lower. ⁸ PM ₁₀ : 30 µg/m ³ or lower. ⁹	1
2	PM _{2.5} : 10 µg/m ³ or lower. ⁹ PM ₁₀ : 20 µg/m ³ or lower. ⁹	2

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 MEET ENHANCED THRESHOLDS FOR ORGANIC GASES (MAX : 1 PT)

For All Spaces:

The following thresholds are met in occupiable spaces:

- a.
- Acetaldehyde: 140 µg/m³ or lower.¹⁰
- b. One of the following:
1. Acrylonitrile: 5 µg/m³ or lower.¹⁰
 2. Caprolactam: 2.2 µg/m³ or lower.¹⁰
- c. Benzene: 3 µg/m³ or lower.¹⁰
- d. Formaldehyde: 9 µg/m³ or lower.¹⁰
- e. Naphthalene: 9 µg/m³ or lower.¹⁰
- f. Toluene: 300 µg/m³ or lower.¹⁰

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 3 MEET ENHANCED THRESHOLDS FOR INORGANIC GASES (MAX : 1 PT)

For All Spaces:

The following thresholds are met:

- a. Carbon monoxide: 7 mg/m³ [6 ppm] or lower.¹¹
- b. Nitrogen dioxide: 40 µg/m³ [21 ppb] or lower.¹¹

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

REFERENCES

1. Porta D, Narduzzi S, Badaloni C, et al. Air pollution and cognitive development at age 7 in a prospective Italian birth cohort. *Epidemiology*. 2016;27(2):228-236. doi:10.1097/EDE.0000000000000405
2. U.S. Environmental Protection Agency. Highlights from the Clean Air Act 40th Anniversary. <https://www.epa.gov/clean-air-act-overview/highlights-clean-air-act-40th-anniversary>.
3. Horr, A., Kaushik, A., Mazroei, A., Katafygiotou, A. & Elsarrag E. Occupant productivity and office indoor environment quality : a review of the literature Occupant Productivity and Office Indoor Environment Quality : A Review of the Literature. 2016. http://usir.salford.ac.uk/39106/3/BAE-D-16-00533_final%2520manuscript%5B1%5D.pdf.
4. Dorgan, CE, Dorgan C. Assessment of link between productivity and indoor air quality. In: *Creating the Productive Workplace*. ; 2005. <https://www.taylorfrancis.com/books/e/9781134265978/chapters/10.4324%2F9780203696880-20>.
5. Calderón-Garcidueñas L, Torres-Jardón R, Kulesza RJ, Park S Bin, D'Angiulli A. Air pollution and detrimental effects on children's brain. The need for a multidisciplinary approach to the issue complexity and challenges. *Front Hum Neurosci*. 2014;8(AUG):1-7. doi:10.3389/fnhum.2014.00613
6. Laumbach RJ, Kipen HM. Respiratory health effects of air pollution: Update on biomass smoke and traffic pollution. *J Allergy Clin Immunol*. 2012;129(1):3-11. doi:10.1016/j.jaci.2011.11.021
7. The Organisation for Economic Co-operation and Development. *OECD Environmental Outlook to 2050: The Consequences of Inaction*. Paris, France: OECD Publishing; 2012. <https://www.naturvardsverket.se/upload/miljoarbete-i-samhallet/internationalt-miljoarbete/multilateralt/oecd/outlook-2050-oecd.pdf>.
8. U.S. Environmental Protection Agency. NAAQS Table. <https://www.epa.gov/criteria-air-pollutants/naaqs-table>.
9. World Health Organization. *Air Quality Guidelines for Particulate Matter, Ozone, Nitrogen Dioxide and Sulfur Dioxide*. Geneva, Switzerland; 2005. http://apps.who.int/iris/bitstream/handle/10665/69477/WHO_SDE_PHE_OEH_06.02_eng.pdf;jsessionid=CD44465Fsequence=1.
10. California Office of Environmental Health Hazard Assessment. *Acute, 8-hour and Chronic Reference Exposure Level (REL) Summary*. <https://oehha.ca.gov/air/general-info/oehha-acute-8-hour-and-chronic-reference-exposure-level-rel-summary>. Published 2019.
11. World Health Organization. *WHO Guidelines for Indoor Air Quality – Selected Pollutants*. Geneva, Switzerland; 2010. http://www.euro.who.int/__data/assets/pdf_file/0009/128169/e94535.pdf.

A06 ENHANCED VENTILATION DESIGN | O (MAX : 3 PT)

Intent : Expel internally generated pollutants and improve air quality in the breathing zone through an increased supply of outdoor air or increased ventilation efficiency.

Summary : This WELL feature requires implementation of advanced ventilation strategies that can achieve higher air quality levels and thus benefit human health and productivity.

Issue : The majority of ventilation standards specify ventilation rates and other measures intended to provide indoor air quality that is merely "acceptable" to building users and reduces the risk of adverse health effects.¹ Even with proper ventilation designed to meet ventilation standards, the concentration of indoor pollutants can exceed concentrations found in outdoor air.^{2,3} Ventilation rates less than 10 L/s per person in all building types are associated with negative perception of air quality and actual health outcomes.⁴

Solutions : Scientific research suggests that an airflow rate significantly exceeding that recommended by standards is needed to minimize sick building syndrome symptoms and to improve human performance and productivity.^{5,6} Since it is difficult to test for every potential pollutant, and because carbon dioxide (CO₂) is easy to detect, CO₂ levels serve as a proxy for other indoor pollutants. A number of CO₂ studies suggest that the risk of sick building syndrome symptoms decreases significantly, when CO₂ concentrations are less than 800 ppm.⁴ One method for decreasing the CO₂ concentration experienced by occupants while minimizing additional energy use is demand-controlled ventilation, in which the delivery rate for outside air is directly linked to the measured CO₂ levels within the space.⁷ Displacement ventilation improves indoor air quality, by delivering fresh outdoor air at the floor level, leaving warmer polluted indoor air, such as CO₂ to be extracted above the height of the ventilation zone.⁸

PART 1 INCREASE OUTDOOR AIR SUPPLY (MAX : 2 PT)

For All Spaces:

Option 1: Increased air supply

For mechanically ventilated buildings, the following requirement is met in all occupiable spaces:

- Exceed outdoor air supply rates described in one of the ventilation guidelines listed in Feature A03 Part 1 by the percentages shown in the table below:

Tier	Thresholds	Points
1	30%	1
2	60%	2

OR

Option 2: Demand control ventilation

For mechanically ventilated buildings, the following requirements are met in at least 90% of regularly occupied spaces:

- A demand-controlled ventilation (DCV) system regulates the outdoor air ventilation rate to keep CO₂ levels less than the thresholds specified in the table below, at the maximum intended occupancy:

Tier	Threshold	Threshold	Points
1	900 ppm	OR 500 ppm above outdoor levels	1
2	750 ppm	OR 350 ppm above outdoor levels	2

- Carbon dioxide is measured at the return air diffusers or in the breathing zone at least 1 m away from doors, windows, air supply diffusers or occupants. At least one sensor is used for each occupancy zone (or per air handling unit, if a single zone is served by multiple air handling units). If the occupancy density/pattern/usage is substantially different in two adjacent areas, each area must be considered a separate zone.

OR

Option 3: Enhanced natural ventilation

For naturally ventilated buildings, the following requirement is met:

- Implement an engineered natural ventilation system that is sufficient to keep CO₂ levels in the breathing zone of all regularly occupied spaces below the specified thresholds at the maximum intended occupancies:

Tier	Threshold	Threshold	Points
1	900 ppm	OR 500 ppm above outdoor levels	1
2	750 ppm	OR 350 ppm above outdoor levels	2

OR

Option 4: Ventilation monitoring

One of the following carbon dioxide thresholds is met in occupiable spaces:

- 750 ppm or less.
- Not more than 350 ppm above outdoor levels.

Note : This Option is worth 2 points. Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

Note : Mixed-mode projects must select the option corresponding to the ventilation mode (mechanical and natural ventilation) in operation for the majority of operating hours within the year.

PART 2 IMPROVE VENTILATION EFFECTIVENESS (MAX : 1 PT)

For All Spaces:

Option 1: Displacement ventilation system

The project uses a displacement ventilation system in at least 90% of regularly occupied spaces, with one of the following as a basis for design:

- a. ASHRAE Guidelines RP-949.⁹
- b. ASHRAE 62.1-2019, "Stratified Air Distribution Systems (Section 6.2.1.2.1).¹⁰
- c. REHVA Guidebook No. 01 (Displacement Ventilation in non-industrial premises).¹¹

OR

Option 2: Personalized ventilation system

At least 50% of workstations incorporate personalized ventilation systems that meet the following requirements:

- a. One of the following strategies is implemented to efficiently deliver air to occupants:
 1. Outdoor air is supplied in the breathing zone through vents incorporated into or mounted on furniture, and the return air diffusers within the space are located at least 2.8 m above the floor.¹⁰
 2. The system's zone air distribution effectiveness (E_z) is at least 1.20.¹⁰
- b. One of the following strategies is implemented to support thermal comfort:
 1. Outdoor air is supplied with an air speed of no greater than 0.25 m/s at the occupant's head.¹⁰
 2. The air speed from each outdoor air supply vent is individually controllable by occupants through a controller within the space, a mobile application or website.

REFERENCES

1. Allen JG, Bernstein A, Cao X, et al. The 9 Foundations of a Healthy Building. Sch Public Heal. 2017:35.
2. Parthasarathy, Srinandini, William J. Fisk TEM. Effect Of Ventilation On Chronic Health Risks In Schools And Offices. Berkeley, CA; 2013. <https://eta.lbl.gov/sites/all/files/publications/lbnl-6121e.pdf>.
3. Shendell DG, Winer AM, Weker R, Colome SD. Evidence of inadequate ventilation in portable classrooms: Results of a pilot study in Los Angeles County. Indoor Air. 2004;14(3):154-158. doi:10.1111/j.1600-0668.2004.00235.x
4. Seppänen OA. Association of ventilation rates and CO₂ concentrations with health and other responses in commercial and institutional buildings. Indoor Air. 1999;9(4):226-252. doi:10.1111/j.1600-0668.1999.00003.x
5. Wyon DP. The effects of indoor air quality on performance and productivity. Indoor Air, Suppl. 2004;14(SUPPL. 7):92-101. doi:10.1111/j.1600-0668.2004.00278.x
6. Sundell J, Levin H, Nazaroff WW, et al. Ventilation rates and health: Multidisciplinary review of the scientific literature. Indoor Air. 2011;21(3):191-204. doi:10.1111/j.1600-0668.2010.00703.x
7. U.S. Department of Energy. Demand Control Ventilation. 2012.
8. Wang Y, Kuckelkorn J, Zhao FY, Spliethoff H. Indoor environment of a classroom in a passive school building with displacement ventilation. Proc BS 2013 13th Conf Int Build Perform Simul Assoc. 2013:1902-1909.
9. American Society of Heating Refrigerating and Air-Conditioning Engineers. Performance Evaluation and Development of Design Guidelines for Displacement Ventilation, RP-949.; 1998.
10. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 62.1-2019: Ventilation for Acceptable Indoor Air Quality. 2019.
11. Skistad H, Mundt E, Nielsen P V., Hagstrom K, Railio J. Displacement Ventilation in Non-Industrial Premises. REHVA Guidebook No. 1. Rehva; 2002.

A07 OPERABLE WINDOWS | O (MAX : 2 PT)

Intent : Increase the supply of high-quality outdoor air and promote a connection to the outdoor environment, by encouraging building users to open windows when outdoor air quality is acceptable.

Summary : This WELL feature requires buildings with operable windows to increase the supply of high-quality outdoor air and promote a connection to the outdoor environment, by encouraging building users to open windows when outdoor air quality is acceptable.

Issue : Inhalation of harmful indoor substances is correlated with adverse health outcomes, such as infectious diseases, cancer, asthma, allergies and other respiratory infections.^{1,2} When outdoor air quality is favorable, opening windows can provide a supply of outdoor air and lower levels of indoor air contaminants. Achieving natural ventilation through open windows can provide a positive human experience.³ However, it is challenging to the ability to maintain strict control over interior air quality. Despite often having conditions of temperature and ventilation below recommended standards, studies have shown that naturally ventilated buildings have fewer people reporting sick building syndrome symptoms than mechanically conditioned buildings.^{4,5} In addition, studies have shown that there is a 7.7% productivity benefit associated with naturally ventilated spaces.^{5,6}

Solutions : When weather and local outdoor parameters indicate high-quality outdoor air, people should be encouraged to make use of natural ventilation strategies. However, ventilating through windows generally introduces more outdoor pollution than mechanical systems with filters.⁷ Human behavior is rated as one of the top three components affecting the generation and persistence of indoor air pollutants.⁷ Educating and informing building users when outdoor conditions are favorable for window opening, can reduce the generation and persistence of indoor air pollutants.

PART 1 PROVIDE OPERABLE WINDOWS (MAX : 1 PT)

Note :

Projects where the average outdoor PM_{2.5} level is 35 µg/m³ or higher may only achieve Part 1: Provide Operable Windows in conjunction with Part 2: Manage Window Use.

For All Spaces:

Project meets one of the below:

- At least 75% of the regularly occupied spaces have operable windows that provide access to outdoor air.
- For each floor, the openable window area is at least 4% the area of the indoor occupiable space.

Note : The World Health Organization's Global Urban Ambient Air Pollution Database may be consulted to view outdoor air quality levels, available at <https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database>

PART 2 MANAGE WINDOW USE (MAX : 1 PT)

Note :

Projects may only achieve this part if Part 1 is also achieved.

For All Spaces:

1: Outdoor air measurement

The following requirement is met:

- Outdoor levels of PM_{2.5}, temperature and humidity are monitored at intervals of at least once per hour, based on a data-gathering station located within 4 km of the building. This monitoring system may be operated by the project or by another entity (e.g., a government).

2: Window operation

Indicator lights and/or digital displays at windows (at least one per room with windows) cue occupants when conditions outside are suitable for opening windows:

- PM_{2.5}: 15 µg/m³ or lower.
- Dry-bulb temperature: within 8 °C of indoor air temperature setpoint.
- Relative Humidity: 65% or lower.

REFERENCES

- Douwes J, Thorne P, Pearce N, Heederik D. Bioaerosol health effects and exposure assessment: Progress and prospects. *Ann Occup Hyg*. 2003;47(3):187-200. doi:10.1093/annhyg/meg032
- Selgrade MK, Plopper CG, Gilmour MI, Conolly RB, Foos BSP. Assessing the health effects and risks associated with children's inhalation exposures - Asthma and allergy. *J Toxicol Environ Heal - Part A Curr Issues*. 2008;71(3):196-207. doi:10.1080/15287390701597897
- Brager GS. Benefits of Improving Occupant Comfort and Well-being in Buildings. In: *Proceedings of the 4th International Holcim Forum for Sustainable Construction: The Economy of Sustainable Construction*. Mumbai; 2013:181-194. https://src.lafargeholcim-foundation.org/dnl/93603859-d59e-498a-b056-405d16e39171/F13_OrangeWS_Brager.pdf.
- Burge S, Hedge A, Wilson S, Bass JH, Robertson A. Sick building syndrome: A study of 4373 office workers. *Ann Occup Hyg*. 1987;31(4 A):493-504. doi:10.1093/annhyg/31.4A.493
- Horro, A., Kaushik, A., Mazroei, A., Katafygiotou, A. & Elsarrag E. Occupant productivity and office indoor environment quality : a review of the literature Occupant Productivity and Office Indoor Environment Quality : A Review of the Literature. 2016. <http://usir.salford.ac.uk/39106/3/BAE-D-16->

00533_final%2520manuscript%5B1%5D.pdf.

6. Wargocki P, Wyon DP, Fanger PO. Productivity is affected by the air quality in offices. *Proc Heal Build 2000*. 2000;1:635-640.
7. Nazaroff WW. Exploring the consequences of climate change for indoor air quality. *Environ Res Lett*. 2013. doi:10.1088/1748-9326/8/1/015022
8. American Society of Heating Refrigerating and Air-Conditioning Engineers. *ASHRAE 62.1-2010: Ventilation for Acceptable Indoor Air Quality*.; 2010.

A08 AIR QUALITY MONITORING AND AWARENESS | O (MAX : 2 PT)

Intent : Monitor indoor air quality issues, as well as inform and educate individuals on the quality of the indoor environment.

Summary : This WELL feature requires the ongoing measurement of contaminant data to educate and empower occupants about their environmental quality.

Issue : Types and concentrations of indoor pollutants continuously fluctuate in any indoor or outdoor environment. For example, cooking in the home can lead to a rapid spike in indoor air pollution.¹ Urban rush hours and waste-burning cause spikes in air pollution outdoors, which can directly impact indoor air quality. Some indoor air pollutants are recognized by their immediate impacts on our body, such as throat irritation or watery eyes.^{2,3} However others that go undetected, are not necessarily benign. According to the U.S. Environmental Protection Agency, some health impacts like respiratory diseases, heart disease and cancer can appear years after exposure.⁴

Solutions : Due to air quality fluctuations, it is important to install air quality sensors and detectors in every building. Because air quality can fluctuate throughout the day in every building, real-time monitoring is necessary to promptly fix any deviations in indoor quality metrics and minimize occupant exposure to pollutants. In addition to having robust and calibrated sensors, positioning them correctly plays a crucial role in the accurate assessment of air quality. Furthermore, educating occupants about the risks associated with elevated air pollutant exposures, along with actions they can take to reduce these risks, can encourage personal agency to seek out opportunities to further curb indoor pollution levels.

PART 1 INSTALL INDOOR AIR MONITORS (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

1: Sensor requirements

The project deploys monitors with sensors that measure at least three of the following parameters in occupiable spaces in compliance with the requirements outlined in the Continuous Monitoring Protocols of the Performance Verification Guidebook:

- a. PM_{2.5} or PM₁₀.
- b. Carbon dioxide.
- c. Carbon monoxide.
- d. Ozone.
- e. Nitrogen dioxide.
- f. Total VOCs.
- g. Formaldehyde.

2: Reporting & maintenance

The following requirement is met:

- a. Proof of calibration or replacement is submitted annually in accordance with the requirements of the WELL Performance Verification Guidebook.

For Dwelling Units & Guest Rooms:

1: Sensor requirements

The project deploys monitors with sensors that measure at least three of the following parameters in occupiable spaces in compliance with the requirements outlined in the Continuous Monitoring Protocols of the Performance Verification Guidebook:

- a. PM_{2.5} or PM₁₀.
- b. Carbon dioxide.
- c. Carbon monoxide.
- d. Ozone.
- e. Nitrogen dioxide.
- f. Total VOCs.
- g. Formaldehyde.

Note :

Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

PART 2 PROMOTE AIR QUALITY AWARENESS (MAX : 1 PT)

For All Spaces:

Note : Projects may only receive points for this part, if Part 1 is also achieved

The air quality data measured in Part 1 of this feature is made available to occupants as follows:

- a. Data are presented through one of the following:
 1. Display screens prominently positioned at a height of 1.1–1.7 m with at least one display per 500 m² of regularly occupied space.
 2. Hosted on a website or phone application accessible to occupants. Signs are present indicating where the data

may be accessed at a density of at least one sign per 500 m² of regularly occupied space.

- b. Data presented include one of the following:
 - 1. Concentrations of the parameters measured.
 - 2. Qualitative results of air quality (e.g., colored-coded levels).
- c. Data are updated at least once every 15 minutes.

REFERENCES

- 1. Wallace LA, Emmerich SJ, Howard-Reed C. Source Strengths of Ultrafine and Fine Particles Due to Cooking with a Gas Stove. *Environ Sci Technol*. 2004;38(8):2304-2311. doi:10.1021/es0306260
- 2. Joshi S. The sick building syndrome. *Indian J Occup Environ Med*. 2008;12(2):61. doi:10.4103/0019-5278.43262
- 3. Selgrade MK, Plopper CG, Gilmour MI, Conolly RB, Foos BSP. Assessing the health effects and risks associated with children's inhalation exposures - Asthma and allergy. *J Toxicol Environ Heal - Part A Curr Issues*. 2008;71(3):196-207. doi:10.1080/15287390701597897
- 4. U.S. Environmental Protection Agency. Introduction to Indoor Air Quality. <https://www.epa.gov/indoor-air-quality-iaq/introduction-indoor-air-quality>.

A09 POLLUTION INFILTRATION MANAGEMENT | O (MAX : 2 PT)

Intent : Minimize the introduction of pollutants into indoor air through the building envelope and at building entrances.

Summary : This WELL feature requires projects to reduce transmission of air and pollutants from outdoors to indoors through the building envelope and entrance.

Issue : Research shows that approximately 65% of outdoor air particle inhalation occurs while indoors.¹ Exposure to high levels of coarse and fine particulate matter inadvertently introduced into the space can lead to respiratory irritation. It has been associated with increases in lung cancer, cardiovascular disease and mortality.² Indoor air quality and thermal comfort can be compromised by leaks and gaps that break the building's air barrier. These weak points are not only wasteful from an energy point of view but can also lead to conditions conducive to mold growth and to the infiltration of pests or polluted air. In addition, building users can introduce particulate matter indoors through their clothes and shoes, including harmful coliforms and *Escherichia coli*, among other toxins.³⁻⁵

Solutions : In addition to building envelope commissioning, there is a need for measures that minimize or prevent the introduction of potentially harmful substances into indoor spaces. An example of such interventions is the installation of entryway walk-off systems and/or entryway air seals at all main building entrances.⁶

PART 1 DESIGN HEALTHY ENTRYWAYS (MAX : 1 PT)

For All Spaces:

1: Building entry design

For any functional building entrance (not including doors to balconies or terraces), the following design features are present:

- a. At least one of the following strategies are in place at entryways that span, at minimum, the width of the entrance and 3 m long in the primary direction of travel (sum of indoor and outdoor length):⁷
 1. Grilles.
 2. Grates or slots.
 3. Rollout mats.
 4. Removable carpet tiles.
 5. Any other product designed to remove dirt from shoes at the entrance.
- b. At least one of the following strategies are in place to slow the movement of air from outdoors to indoors:
 1. Building entry vestibule with two typically closed doorways.
 2. Revolving entrance doors.
 3. Buildings with an entrance that is outside of the project boundary, or buildings with an entrance lobby that is not regularly occupied, must have at least three typically shut doors that separate the outdoors from all regularly occupied spaces within the project boundary.
 4. Air curtains installed and commissioned in accordance with ASHRAE Standard 90.1-2019.

2: Building entry maintenance

Building entryway systems are cleaned, as follows:

- a. Wet-cleaned at least once a week, or as instructed by manufacturer.
- b. Vacuumed at least once a day, or as instructed by manufacturer.

For Outdoor Sport Areas:

The following requirement is met:

- a. All facilities adjacent to an outdoor sports field have an area (e.g., staging area, mudroom, drying room) that separates the playing field from other internal areas to capture moisture and debris.

Note :

Interiors projects may count exterior building entrances that meet the feature requirements, even if they are outside of the project boundary.

PART 2 PERFORM ENVELOPE COMMISSIONING (MAX : 1 PT)

For All Spaces:

For projects undergoing design and construction, the following requirements are met:

- a. The project uses a façade engineer that is responsible for defining the building envelope performance metrics (including materials, components, assemblies and systems) at the concept design stage.
- b. The building envelope performance requirements are included in the Basis of Design document and reflect the Owner's Project Requirements.
- c. The commissioning process includes envelope commissioning for air infiltration and leakage, which is reflected in the specification and commissioning plan.
- d. The envelope commissioning process is executed, as outlined in the commissioning plan.
- e. The envelope commissioning plan is included in the project Operation & Maintenance (O&M) Manual.

Note :

Interiors projects may count commissioning performed on the building envelope, even if outside of the project boundary.

REFERENCES

1. Fisk WJ. Review of some effects of climate change on indoor environmental quality and health and associated no-regrets mitigation measures. *Build Environ*. 2015;86(January):70-80. doi:10.1016/j.buildenv.2014.12.024
2. Du Y, Xu X, Chu M, Guo Y, Wang J. Air particulate matter and cardiovascular disease: The epidemiological, biomedical and clinical evidence. *J Thorac Dis*. 2016;8(1):E8-E19. doi:10.3978/j.issn.2072-1439.2015.11.37
3. Licina D, Tian Y, Nazaroff WW. Emission rates and the personal cloud effect associated with particle release from the perihuman environment. *Indoor Air*. 2017;27(4):791-802. doi:10.1111/ina.12365
4. Lax S, Hampton-Marcell JT, Gibbons SM, et al. Forensic analysis of the microbiome of phones and shoes. *Microbiome*. 2015;3(1):21. doi:10.1186/s40168-015-0082-9
5. Licina D, Nazaroff WW. Clothing as a transport vector for airborne particles: Chamber study. *Indoor Air*. 2018;28(3):404-414. doi:10.1111/ina.12452
6. Younes C, Shdid CA, Bitsuamlak G. Air infiltration through building envelopes: A review. *J Build Phys*. 2012;35(3):267-302. doi:10.1177/1744259111423085
7. U.S. Green Building Council. LEED BD+C: New Construction | v4 - LEED v4. 2018. https://www.usgbc.org/sites/default/files/LEED_v4_BDC_07.2.18_current.pdf.
8. Wenger Corporation. Planning your athletic facility. [https://www.wengercorp.com/Construct/docs/Athletic Planning Guide by Wenger GearBoss.pdf](https://www.wengercorp.com/Construct/docs/Athletic_Planning_Guide_by_Wenger_GearBoss.pdf).

A10 COMBUSTION MINIMIZATION | O (MAX : 1 PT)

Intent : Reduce human exposure to combustion-related air pollution from heating and transportation sources.

Summary : This WELL feature requires projects to utilize low-emission combustion products or eliminate combustion-based products entirely.

Issue : Combustion-related emissions, mostly from space heating, cooking and nearby transportation, are often a major and overlooked source of indoor air pollution.¹ Inefficient heating practices, cooking or other combustion activities produce high levels of indoor air pollution that include a range of health-damaging pollutants, such as fine particles, nitrogen dioxide and carbon monoxide.² In addition to nausea, loss of consciousness and death³, inhalation of elevated levels of carbon monoxide is linked to headaches, visual impairment, reduced cognitive functioning and the ability to perform complex tasks⁴.

Solutions : Opting for non-combustion or low-emission combustion products is an important first step toward the reduction of carbon monoxide, nitrogen dioxide, small particles and other combustion by-products in the air.⁵

PART 1 MANAGE COMBUSTION (MAX : 1 PT)

For All Spaces except Commercial Kitchen Spaces:

1: Appliance and heater combustion ban

The following requirements are met for combustion-based fireplaces, stoves, space heaters, ranges and ovens:

- a. Are not used in indoors.⁶
- b. If located outdoors, only use natural gas / propane and are located at least 3.3 m from the building⁸

2: Low-emission combustion sources

Equipment used by the project for heating, cooling, water heating, process heating or power generation (including back-up, if used for more than 200 hours per year) meets one or more of the following requirements:

- a. Comply with California's South Coast Air Quality Management District emission rules for pollution.⁷
- b. Are electric.
- c. Are supplied by district heating or cooling.

3: Engine exhaust reduction

The following requirement is met:

- a. Vehicle engine idling for more than 30 seconds is prohibited in all pick-up, drop-off and parking areas at the building site controlled by the project. "No idling" signage is present at these locations indicating this rule.

REFERENCES

1. Liu J, Mauzerall DL, Chen Q, et al. Air pollutant emissions from Chinese households: A major and underappreciated ambient pollution source. *Proc Natl Acad Sci*. 2016;113(28):7756-7761. doi:10.1073/pnas.1604537113
2. World Health Organization. Household (Indoor) Air Pollution. <http://www.who.int/airpollution/household/en/>. Published 2016.
3. Weaver LK, Hopkins RO, Chan KJ, et al. Hyperbaric Oxygen for Acute Carbon Monoxide Poisoning. *N Engl J Med*. 2002;347(14):1057-1067. doi:10.1056/NEJMoa013121
4. Centers for Disease Control and Prevention. Carbom Monoxide Poisoning. <https://ephtracking.cdc.gov/showCoRisk.action>.
5. Bruce N, Pope D, Rehfuess E, Balakrishnan K, Adair-Rohani H, Dora C. WHO indoor air quality guidelines on household fuel combustion: Strategy implications of new evidence on interventions and exposure-risk functions. *Atmos Environ*. 2015;106:451-457. doi:10.1016/j.atmosenv.2014.08.064
6. U.S. Environmental Protection Agency. How can you protect your heart from air pollution? <https://www.epa.gov/sciencematters/how-can-you-protect-your-heart-air-pollution>. Published 2018.
7. South Coast Air Quality Management District. SCAMD Rules and Regulations, Regulation XI - Source Specific Standards. <http://www.aqmd.gov/home/rules-compliance/rules/scaqmd-rule-book/regulation-xi>.
8. State of California. California Building Standards Code 2019 Title 24, Part 9, California Fire Code – Section 307 Open Burning, Recreational Fires and Portable Outdoor Fireplaces. <https://www.dgs.ca.gov/BSC/Codes>. Published 2019.

A11 SOURCE SEPARATION | O (MAX : 1 PT)

Intent : Preserve indoor air quality and maximize olfactory comfort in occupied spaces through the isolation and proper ventilation of indoor pollution sources and chemical storage areas.

Summary : This WELL feature requires strategies that isolate key sources of odors, germs, pollution or humidity through doors or dedicated exhaust.

Issue : Air pollution can be created from many indoor sources, including cleaning products, office equipment and humid environments. Chemical storage closets that house cleaning products, can be a source of harmful vapors, including Volatile Organic Compounds (VOCs) that are linked to cancer, organ and central nervous system damage.^{1,2} Copy rooms can contribute to the production of ozone, which has been associated with lung inflammation, chest pain, wheezing, coughing, shortness of breath and the exacerbation of respiratory illnesses, such as pneumonia and asthma.³⁻⁵ Bathrooms can encourage the growth of mold and mildew, which have been associated with allergic reactions and aggravate lung diseases, such as asthma.⁶ Exposure to air pollutants in kitchens, such as particulate matter, VOCs and polycyclic aromatic hydrocarbons, has been associated with kidney inflammation.⁷

Solutions : The most effective way to curb air pollution in buildings is to eliminate individual sources or capture emissions at the source, before they spread to surroundings.⁸ For air pollution sources that are inevitable, physical separation of such sources combined with direct ventilation exhaust systems is an effective means of protecting individuals.

PART 1 MANAGE POLLUTION AND EXHAUST (MAX : 1 PT)

For All Spaces except Dwelling Units & Guest Rooms:

For all bathrooms, kitchens, rooms for cleaning and chemical storage, rooms with high-volume printers and copiers and high-humidity areas, the following requirements are met:

- a. Meet one of the following:
 1. Are separated from all adjacent regularly occupied spaces with self-closing doors and/or vestibules.
 2. Are negatively pressurized compared with adjacent regularly occupied spaces.
- b. Utilize exhaust fans such that the return air is vented outdoors and not recirculated.^{9,10}

For Commercial Kitchen Spaces:

Option 1: Control cooking pollution

The following requirements are met:

- a. Canopy hoods have side or partial panels, when allowable by code.¹¹
- b. Type II hood overhangs and setbacks comply with ASHRAE 154-2011.¹²
- c. The vertical distance between the front lower lip of the hood and the cooking surface is less than or equal to 1.2 m.¹²
- d. Replacement air velocity near (or directed at) the hood is less than 0.38 m/s.¹¹
- e. Replacement air introduced directly into the exhaust hood cavity does not exceed 10% of the hood exhaust airflow rate.¹³
- f. Appliances are grouped under exhaust hoods according to effluent production and associated ventilation requirements, as specified in ASHRAE 154-2011, per hood type (defined by the classifications used in ASHRAE 154-2011 for light, medium, heavy and extra-heavy appliance duty levels).¹²
- g. Appliances have a rear seal between the appliance and the wall, when allowable by code.¹⁵
- h. Appliances located at the end of a cook line requiring exhaust airflow rates greater than 460 L/s/m have a full side panel or an end wall.¹²

OR

Option 2: No applicable commercial appliances

The following requirements are met:

- a. There are no appliances that require a hood, as specified in ASHRAE 154-2011 (Table 2 - Type I Hood Requirements by Appliance Type, Table 2 - Type II Hood Requirements by Appliance Description).¹²

For Dwelling Units & Guest Rooms:

The following requirements are met:

- a. Where cooking burners and stove top cooking appliances are present, a range hood meets the following:
 1. Exhaust air is vented directly to the outdoors.¹⁶
 2. Exhaust air outlets are separated from any air intakes by at least 3 m, unless otherwise specified by local code.¹⁷
 3. The minimum operating exhaust airflow rate is the greater of 150 L/s per linear meter of range hood width or 94 L/s.¹⁸
 4. The range hood device, when in operation, covers at least 75% of the burner area.¹⁹
- b. All bathrooms contain one of the following:

1. Exhaust fan.
2. Operable window.

REFERENCES

1. Nurmatov UB, Tagiyeva N, Semple S, Devereux G, Sheikh A. Volatile organic compounds and risk of asthma and allergy: A systematic review. *Eur Respir Rev.* 2015;24(135):92-101. doi:10.1183/09059180.00000714
2. U.S. Environmental Protection Agency. Volatile Organic Compounds' Impact on Indoor Air Quality. <https://www.epa.gov/indoor-air-quality-iaq/volatile-organic-compounds-impact-indoor-air-quality>.
3. U.S. Environmental Protection Agency. Health Effects of Ozone in Patients with Asthma and Other Chronic Respiratory Disease. <https://www.epa.gov/ozone-pollution-and-your-patients-health/health-effects-ozone-patients-asthma-and-other-chronic>.
4. The National Institute of Environmental Health Sciences. Ozone. <https://www.niehs.nih.gov/health/topics/agents/ozone/index.cfm>. Published 2008.
5. Tuomi T, Engstrom B, Niemela R, Svinhufvud J, Reijula K. Emission of ozone and organic volatiles from a selection of laser printers and photocopiers. *Appl Occup Environ Hyg.* 2000;15(8):629-634. doi:10.1080/10473220050075635
6. Mendell MJ, Mirer AG, Cheung K, Tong M, Douwes J. Respiratory and allergic health effects of dampness, mold, and dampness-related agents: A review of the epidemiologic evidence. *Environ Health Perspect.* 2011;119(6):748-756. doi:10.1289/ehp.1002410
7. Singh A, Kamal R, Mudiam MKR, et al. Heat and PAHs emissions in indoor kitchen air and its impact on kidney dysfunctions among kitchen workers in Lucknow, North India. Jia Z, ed. *PLoS One.* 2016;11(2):e0148641. doi:10.1371/journal.pone.0148641
8. U.S. Environmental Protection Agency. Improving Indoor Air Quality. <https://www.epa.gov/indoor-air-quality-iaq/improving-indoor-air-quality>.
9. U.S. Green Building Council. LEED BD+C: New Construction | v4 - LEED v4. 2018. https://www.usgbc.org/sites/default/files/LEED_v4_BDC_07.2.18_current.pdf.
10. Wenger Corporation. Planning your athletic facility. <https://www.wengercorp.com/Construct/docs/AthleticPlanningGuidebyWengerGearBoss.pdf>.
11. California Energy Commission. Design Guide Improving Commercial Kitchen Ventilation System Performance.; 2003. http://www.energy.ca.gov/reports/2003-06-13_500-03-034F.PDF.
12. International Codes Council. 2015 Minnesota Mechanical and Fuel Gas Code with the ANSI/ASHRAE Standard 154-2011. Presented at the: 2015. <https://codes.iccsafe.org/public/document/MMFGC2015/copyright>.
13. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 90.1: Energy Standard for Buildings Except Low-Rise Residential Buildings.; 2010. http://www.usalighting.com/stuff/contentmgr/files/1/b90ce247855d0f17438484c003877338/misc/ashrae_90_1_2010.pdf
14. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 189.1: Standard for the Design of High-Performance Green Buildings Except Low-Rise Residential Buildings.; 2011. http://www.ditar.cl/archivos/Normas_ASHRAE/T1160ASHRAE-189.1-2009-Design-H-P-Green-Buildings.pdf.
15. Sobiski P, Swierczyna R, Fisher D. Effects of Range Top Diversity, Range Accessories, and Hood Dimensions on Commercial Kitchen Hood Performance. Vol 112 PART 1. San Francisco, CA; 2006.
16. Militello-Hourigan RE, Miller SL. The impacts of cooking and an assessment of indoor air quality in Colorado passive and tightly constructed homes. *Build Environ.* 2018. doi:10.1016/j.buildenv.2018.08.044
17. American Society of Heating Refrigerating and Air Conditioning Engineers (ASHRAE). ASHRAE 154-2003: Ventilation for Commercial Cooking Operations. 2003. http://sspc154.ashraepcs.org/pdf/STD154_2003.pdf.
18. Home Ventilating Institute. What is a range hood? https://www.hvi.org/HVIORG/document-server/?cfp=HVIORG/assets/File/public/publications/HVIRangeHood_4Feb08.pdf.
19. Singer BC, Delp WW, Apte MG. Experimental evaluation of installed cooking exhaust fan performance. Ernest Orlando Lawrence Berkeley Natl Lab. 2012.

A12 AIR FILTRATION | O (MAX : 1 PT)

Intent : Reduce indoor and outdoor airborne contaminants through air filtration.

Summary : This WELL feature requires projects with mechanically ventilated spaces to implement adequate air filtration and document a maintenance protocol for installed filters.

Issue : Exposure to particulate matter (PM) is associated with many negative health outcomes. PM₁₀ can block and inflame airways, causing a range of respiratory-related conditions that can lead to illness or death.¹ PM_{2.5} poses even greater health risks compared to PM₁₀, because it can penetrate deep into the lungs, enter the bloodstream and as a result, cause a variety of health issues, including heart disease and other cardiovascular complications.¹

Solutions : Selection and installation of adequate media filters is one of the key mechanisms for minimizing exposure to outdoor and indoor air pollution. Studies have shown that decreased exposure to particulate matter by filtration of recirculated indoor air is associated with a reduced risk of cardiovascular disease and is an effective control measure for reducing allergic respiratory disease.^{2,3} In addition, regular filter maintenance is critical to ensure proper air filtration and the efficiency of the air conditioning system. During the operation, filters should be replaced when they become loaded with particles, since they will begin to reduce airflow and increase pressure drop. Overloaded filters not only restrict airflow rate but can also result in a loss of filtration efficiency. It is also critical for projects to be aware whether the building is located in an area with elevated outdoor air pollution, since these projects often need to install a pre-filtration stage, in addition to the primary filtration, to maintain high-quality indoor air.⁴

PART 1 IMPLEMENT PARTICLE FILTRATION (MAX : 1 PT)

For All Spaces:

1: Filtration levels

The following requirement is met:

- a. Media filters are used in the ventilation system to filter outdoor air supplied to the space, in accordance with thresholds specified in the table below:^{5,6}

Annual Average Outdoor PM _{2.5} Threshold	Average Air Filtration Efficiency (particles 0.3-1 µm)
23 µg/m ³ or less	≥ 35% (e.g., MERV 12 or M6)
24–39 µg/m ³	≥ 75% (e.g., MERV 14, F8 or ePM1 75%)
40 µg/m ³ or greater	≥ 95% (e.g., MERV 16, E10 or ePM1 95%)

2: Filter maintenance

The following requirement is met:

- a. Evidence that the filter has been replaced according to the manufacturer's recommendation is submitted annually through the WELL digital platform.

Note :

The World Health Organization's Global Urban Ambient Air Pollution Database may be consulted to view outdoor air quality levels, available at <https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database>

REFERENCES

1. World Health Organization. Health Effects of Particulate Matter: Policy Implications for Countries in Eastern Europe, Caucasus and Central Asia. Geneva, Switzerland <http://www.euro.who.int/en/health-topics/environment-and-health/air-quality/publications/2013/health-effects-of-particulate-matter.-policy-implications-for-countries-in-eastern-europe,-caucasus-and-central-asia-2013>.
2. Bräuner EV, Forchhammer L, Møller P, et al. Indoor particles affect vascular function in the aged: An air filtration-based intervention study. *Am J Respir Crit Care Med*. 2008;177(4):419-425. doi:10.1164/rccm.200704-632OC
3. Sublett JL. Effectiveness of air filters and air cleaners in allergic respiratory diseases: A review of the recent literature. *Curr Allergy Asthma Rep*. 2011;11(5):395-402. doi:10.1007/s11882-011-0208-5
4. Centers for Disease Control and Prevention. Filtration and Air-Cleaning Systems to Protect Building Environments from Airborne Chemical, Biological, or Radiological Attacks. <https://www.cdc.gov/niosh/docs/2003-136/default.html>. Published 2003.
5. Stephens B, Brennan T, Harriman L. Selecting Ventilation Air Filters to Reduce PM 2.5 Of Outdoor Origin. ASHRAE J. 2016. http://www.conforlab.com.br/wp-content/uploads/2016/10/2016Sep_012-021_HarrimanFiltersToReducePM2.5.pdf.
6. Camfil. Comparison Chart ASHRAE 52.2, 52.1, EN779, EN1882. [https://www.camfil.com/FileArchive/Industries/Gas turbines and other power systems/Filter brochures/Filter_class_chart_ASHRAE_EN2012.pdf](https://www.camfil.com/FileArchive/Industries/Gas%20turbines%20and%20other%20power%20systems/Filter%20brochures/Filter_class_chart_ASHRAE_EN2012.pdf). Accessed April 1, 2018.

A13 ENHANCED SUPPLY AIR | O (MAX : 3 PT)

Intent : Mitigate risks from indoor contamination and pollution sources, such as infectious disease particles and volatile organic compounds (VOC).

Summary : This WELL feature requires the projects to use supply air that is not recirculated or that is treated with carbon filters, media filters and/or Ultraviolet Germicidal Irradiation (UVGI).

Issue : Building materials, furnishings (e.g., carpets and furniture finishes), fabrics, cleaning products, personal care products, adhesives, solvents and air fresheners can all emit VOCs or semi-volatile organic compounds (SVOCs) into the indoor environment.^{1,2} VOCs include benzene, formaldehyde and other chemical compounds which, at high concentrations, can lead to the irritation of the nose and pharynx and have been associated with leukemia and nasopharyngeal cancer.^{3,4} Health effects can also include damage to the liver, kidneys and central nervous system.⁵ When humans exhale, they can release pathogens, such as COVID-19 or Influenza particles, which can remain suspended in the air for several hours or longer. These pathogens can be recirculated through ducts and HVAC systems of buildings, increasing the risk of transmission of disease indoors.⁶⁻⁸

Solutions : The simplest way to avoid recirculating contaminated air is to not recirculate it by supplying spaces with 100% outdoor air. Unfortunately, in certain climates this can result in high energy usage, although this can be mitigated through the use of heat recovery systems.⁹ In buildings where recirculated air is utilized, it can be treated to remove contaminants. Carbon is capable of filtering VOCs and ozone from the air that passes through.^{10,11} HEPA or near-HEPA filters can help remove virus particles, since the virus often travels as an attachment to a larger particle.^{12,13} UVGI systems can also be effective, both when irradiating the upper portion of the room or when installed in the air ducts, so long as they are powerful and/or the air speed is slow enough to provide a sufficient UV dose.^{13,14} Other technologies exist, but should be evaluated carefully. For example, more novel devices that release oxidants into rooms have less robust evidence for safety and efficacy outside of laboratory conditions.^{15,16} Finally, in-room air purifiers can be beneficial because the clean air is often provided within the breathing zone. For optimal performance, air filtration systems need to be maintained according to the manufacturer's instructions.

PART 1 IMPROVE SUPPLY AIR (MAX : 1 PT)

For All Spaces:

Option 1: 100% Outdoor air provisions

The following requirements are met:

- a. Each regularly occupied space is ventilated with outdoor air that has not recirculated from other rooms within the building.

OR

Option 2: Cleaning and purification devices

All regularly occupied spaces with recirculated air are treated with purification/cleaning system(s), either in the HVAC system or as a standalone device, which meet the following requirements:

- a. Use, at a minimum, two of the following air purification/cleaning technologies:
 1. Activated carbon filter.
 2. Media filter with an average removal efficiency of $\geq 75\%$ for particles 0.3-1 μm in size (e.g., MERV 14; F8; ePM1 75%).
 3. UVGI to treat the moving air OR upper-room UVGI.
- b. Comply with one of the following:
 1. Are validated under UL 2998 Zero Ozone Emissions Validation or Intertek Zero Ozone Verification.¹⁵
 2. Do not use electronic air cleaners.
- c. Are not designed to release ions, reactants, or other molecules into occupiable spaces to disinfect or clean the air.¹⁶
- d. Are sized appropriately to the room volume or area, based on manufacturer specifications.
- e. Are maintained according to the manufacturer's recommendations. Documentation of maintenance is submitted annually through the WELL digital platform.

PART 2 B PROVIDE CLEAN AIRFLOW RATES FOR CONTROL OF INFECTIOUS AEROSOLS (MAX : 2 PT)

For All Spaces:

Option 1: Regular Airflow Operation

The mechanical system provides all regularly occupied spaces with clean airflow rates for infection risk management as set in one of the following guidelines:

- a. ASHRAE 241-2023 Table 5-1 Equivalent Clean Airflow Rates (Section 6 - Clean Airflow Rate Equation or Appendix C - In-place Modeling Method).¹⁷
- b. 5 Equivalent Air Changes per hour (5 e/ACH) calculated in accordance with Appendix A1.¹⁸

OR

Option 2: Operation Under Infectious Disease Management Mode

The following requirements are met:

- a. The mechanical system is capable of providing all regularly occupied spaces with clean airflow rates for infection risk management as set in one of the following guidelines:
 1. ASHRAE 241-2023 Table 5-1 Equivalent Clean Airflow Rates (Section 6 - Clean Airflow Rate Equation or Appendix C - In-place Modeling Method).¹⁷
 2. 5 Equivalent Air Changes per hour (5 e/ACH) calculated in accordance with Appendix A1.¹⁸

- b. The project has an operational plan for Infection Risk Management Mode (IRMM) that includes the following:¹⁷
1. The conditions under which the IRMM will be activated.
 2. The engineering controls (e.g., changes in ventilation or air treatment) and non-engineering controls (e.g., changes in occupancy) that will be utilized to achieve the target clean airflow rates.
 3. The operational and maintenance procedures in place to implement target clean airflow rates within 48-hours of the decision to activate IRMM.
 4. An inventory of HVAC system consumables (e.g., filters, adsorption media, UV bulbs) that is updated when significant equipment modifications have been implemented or after replacement consumables have been utilized.
 5. A requirement that there is at least one set of replacement consumables available to install during IRMM.

Note :

This feature is a beta strategy and has an additional documentation requirement (beta feature feedback form). The feedback form supports IWBI in developing new features that are effective and applicable to projects around the world.

REFERENCES

1. Salthammer T. Emissions of Volatile Organic Compounds from Products and Materials in Indoor Environments. *Handb Environ Chem*. 2004;4:37-71. doi:10.1007/b94830
2. Wallace LA, Pellizzari E, Leaderer B, Zelon H, Sheldon L. Emissions of volatile organic compounds from building materials and consumer products. *Atmos Environ*. 1987;21(2):385-393. doi:10.1016/0004-6981(87)90017-5
3. Nurmatov UB, Tagiyeva N, Semple S, Devereux G, Sheikh A. Volatile organic compounds and risk of asthma and allergy: A systematic review. *Eur Respir Rev*. 2015;24(135):92-101. doi:10.1183/09059180.00000714
4. Nielsen GD, Larsen ST, Wolkoff P. Re-evaluation of the WHO (2010) formaldehyde indoor air quality guideline for cancer risk assessment. *Arch Toxicol*. 2017;91(1):35-61. doi:10.1007/s00204-016-1733-8
5. U.S. Environmental Protection Agency. Volatile Organic Compounds' Impact on Indoor Air Quality. <https://www.epa.gov/indoor-air-quality-iaq/volatile-organic-compounds-impact-indoor-air-quality>.
6. Stariolo DA. COVID-19 in Air Suspensions.; 2020. <https://arxiv.org/abs/2004.05699>. Accessed June 15, 2020.
7. Federation of European Heating Ventilation and Air-Conditioning Associations. How to operate and use building services in order to prevent the spread of the coronavirus disease (COVID-19) virus (SARS-CoV-2) in workplaces. 2020.
8. Dietz L, Horve PF, Coil DA, Fretz M, Eisen JA, Van Den Wymelenberg K. 2019 Novel Coronavirus (COVID-19) Pandemic: Built Environment Considerations To Reduce Transmission. Gilbert JA, ed. *mSystems*. 2020;5(2):e00245-20. doi:10.1128/mSystems.00245-20
9. Zemitis J, Borodinecs A. Energy saving potential of ventilation systems with exhaust air heat recovery. *IOP Conf Ser Mater Sci Eng*. 2019;660:012019. doi:10.1088/1757-899X/660/1/012019
10. Fisk WJ, Spears M, Sullivan DP, Mendell M. Ozone Removal by Filters Containing Activated Carbon: A Pilot Study.; 2009. <https://www.osti.gov/scitech/servlets/purl/1050670/>.
11. Sekine Y, Fukuda M, Takao Y, Ozano T, Sakuramoto H, Wang KW. Simultaneous removal of formaldehyde and benzene in indoor air with a combination of sorption- and decomposition-type air filters. *Environ Technol*. 2011;33(15-16):1983-1989. doi:10.1080/09593330.2011.562924
12. Liu Y, Ning Z, Chen Y, et al. Aerodynamic analysis of SARS-CoV-2 in two Wuhan hospitals. *Nature*. 2020. doi:10.1038/s41586-020-2271-3
13. The American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE Position Document on Infectious Aerosols. 2020.
14. Yang Y, Zhang H, Nunayon SS, Chan V, Lai AC. Disinfection efficacy of ultraviolet germicidal irradiation on airborne bacteria in ventilation ducts. *Indoor Air*. 2018;28(6):806-817. doi:10.1111/ina.12504
15. U.S. Environmental Protection Agency. Air Cleaners, HVAC Filters, and Coronavirus (COVID-19). Accessed: October 7, 2022. [Reference](#)
16. ASHRAE. ASHRAE Position Document on Filtration and Air Cleaning. Published: 2015. [Reference](#)
17. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 241-2023: Control of Infectious Aerosols.; 2023.
18. Centers for Disease Control and Prevention. How Much Ventilation is Enough? [Reference](#).

A14 MICROBE AND MOLD CONTROL | O (MAX : 1 PT)

Intent : Reduce mold and bacteria growth within the building mechanical system.

Summary : This WELL feature requires projects to utilize UVGI systems and/or conduct regular inspections on components of the cooling system to reduce or eliminate growth of microbes and mold.

Issue : Mold will grow in places with an acceptable temperature range, a nutrient source and sufficient moisture.¹ Air conditioner systems, specifically cooling coils, have high levels of moisture condensation and serve as the perfect breeding grounds for mold.² Mold developed on cooling coils may shed particles into the building's indoor air and trigger asthma, headaches, allergies and other respiratory system disorders.³⁻⁶ Exposure to mold has also been associated with hypersensitivity pneumonitis, allergic rhinitis, bronchitis, lung tumor development, eczema and toxic mold syndrome.^{5,7}

Solutions : The growth of microorganisms and mold can be prevented and mitigated through a combination of regular mold inspections and placement of microbe inactivation techniques, such as Ultraviolet Germicidal Irradiation (UVGI) systems. Periodic inspections and maintenance of cooling systems are good preventative methods to reduce system failures that contribute to an increased potential for mold proliferation. Expenses associated with proper maintenance will often offset typical costs associated with mold remediation.² UVGI has been shown to dramatically reduce mold and bacteria growth on cooling coils and destroy microbial films that accumulate on their surfaces.⁸ Studies have also associated the implementation of UVGI systems with a simultaneous reduction in viable microorganisms and respiratory disorders in the workplace.⁹

PART 1 IMPLEMENT ULTRAVIOLET TREATMENT FOR HVAC SURFACES (MAX : 1 PT)

For All Spaces:

1: UV system design

The following requirements are met:

- a. All central air handling units use ultraviolet lamps to irradiate the surfaces of the cooling coils and drain pans.¹⁰
- b. All cooling coils and drain pans not associated with central air handling units (e.g., those used in fan coil units or supplementary air handling units) either:
 1. Are irradiated by ultraviolet lamps.
 2. May be opened for inspection for mold growth and cleaned, if necessary.

2: UV system maintenance and inspection

The following requirements are met:

- a. Evidence that the ultraviolet lamps have been replaced or maintained, according to manufacturer's recommendation is submitted annually through the WELL digital platform.
- b. All cooling coils without ultraviolet lamps (if applicable) are inspected on a quarterly basis for mold growth and cleaned if necessary. Dated photos demonstrating adherence are submitted annually through the WELL digital platform.

REFERENCES

1. Air Conditioning Heating & Refrigeration News. Reducing Mold Growth in HVAC Equipment. <https://www.enviro-tec.com/pdf/catalog/WP-ReducingMoldGrowth.pdf>. Published 2003.
2. Fink R. Mold/Bacteria Protection of A/C Coil. ACHR News. <https://www.achrnews.com/articles/106240-mold-bacteria-protection-of-a-c-coil>. Published 2008.
3. Zock J-P, Jarvis D, Luczynska C, Sunyer J, Burney P, European Community Respiratory Health Survey. Housing characteristics, reported mold exposure, and asthma in the European Community Respiratory Health Survey. J Allergy Clin Immunol. 2002;110(2):285-292. [https://www.jacionline.org/article/S0091-6749\(02\)00092-1/pdf](https://www.jacionline.org/article/S0091-6749(02)00092-1/pdf).
4. Mendell MJ, Mirer AG, Cheung K, Tong M, Douwes J. Respiratory and allergic health effects of dampness, mold, and dampness-related agents: A review of the epidemiologic evidence. Environ Health Perspect. 2011;119(6):748-756. doi:10.1289/ehp.1002410
5. Fisk WJ, Eliseeva EA, Mendell MJ. Association of residential dampness and mold with respiratory tract infections and bronchitis: A meta-analysis. Environ Heal A Glob Access Sci Source. 2010;9(1):72. doi:10.1186/1476-069X-9-72
6. Jones R, Recer GM, Hwang SA, Lin S. Association between indoor mold and asthma among children in Buffalo, New York. Indoor Air. 2011;21(2):156-164. doi:10.1111/j.1600-0668.2010.00692.x
7. Pettigrew HD, Selmi CF, Teuber SS, Gershwin ME. Mold and human health: Separating the wheat from the chaff. Clin Rev Allergy Immunol. 2010;38(2-3):148-155. doi:10.1007/s12016-009-8175-5
8. Berkeley Lab, LBNL Indoor Environment Group. Using Ultraviolet Germicidal Lights for Air Cleaning. <https://iaqscience.lbl.gov/air-uv>.
9. Dick Menzies, Julia Popa, James A Hanley, Thomas Rand DKM. Effect of Ultraviolet Germicidal Lights Installed in Office Ventilation Systems on Workers' Health and Wellbeing: Double-Blind Multiple Crossover Trial.
10. The American Society of Heating Refrigerating and Air-Conditioning Engineers. Standard 185.2-2014 -- Method of Testing Ultraviolet Lamps for Use in HVAC&R Units or Air Ducts to Inactivate Microorganisms on Irradiated Surfaces.; 2014. https://www.techstreet.com/standards/ashrae-185-2-2014?product_id=1881838.

APPENDIX A1:

$$eACH = Q / [\text{room volume}]^{18}$$

Where Q = A + B + C + D (total clean volumetric air flow rate)

To determine the total clean volumetric air flow rate (Q) for a room, sum the air flow rates for individual systems (as applicable to the HVAC system) per the calculations below.

A: Ventilation Systems¹⁸

The greater of:

The full mechanical HVAC outdoor air supply rate

The full mechanical HVAC system's exhaust rate, provided the transfer air is from non-regularly occupied spaces or outdoors

B: CADR-rated Media Filters (Standalone Systems)^{17,18}

One of the following:

The full m-CADR rate (AHAM Certified per ANSI/AHAM Standard AC-5) or, if unavailable,

The full smoke-CADR rate (AHAM Certified per ANSI/AHAM Standard AC-1)

C: In-duct or non CADR-rated Media Filters¹⁸

Multiply the air supply rate (e.g., mechanical HVAC fan, blower fan) by the weighted clean air below:

MERV-A (ASHRAE 52.2)	ePM2.5 (ISO 16890)	Weighted Clean Air Factor (MERV E2 [1-3 um] minimum arrestance efficiency)
< 13	≤ 79	0
13	80	85
14-15	85	90
16	90	95
17-19 (HEPA)	99	99

D: In-duct UVGI¹⁷

Multiply the air supply rate by the infectious aerosol reduction efficiency determined in accordance with ANSI/ASHRAE 185.1 with MS2 challenge organism.